An Argument for Gradient Grammaticality

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Any *gradience* observed in acceptability judgement experiments is typically attributed to online sentence-processing mechanisms. By contrast, grammaticality is usually taken to be *binary*: sentences are either grammatical or not. This paper presents experimental evidence against this latter view.

In Experiment 1, with 70 native speakers of English and using the Thermometer Method (Featherston 2008, 2009), we tested syntactic selectional properties of eight English predicates semantically compatible with a propositional complement: *account*, *annoyed*, *ashamed*, *depend*, *familiar*, *speak*, *suffer*, *talk*. Each predicate appeared in 8 tightly controlled item pairs like (1), yielding 64 stimuli distributed across two counterbalanced lists.

(1) a. Justin Bieber can **depend** [$_{\text{PP}}$ on the fact [$_{\text{CP}}$ that his fans still love his early songs]].
b. Justin Bieber can **depend** [$_{\text{CP}}$ that his fans still love his early songs].

As shown in Figure 1a, participants gave uniformly high ratings to PP complements (as in (1a)) expressing propositional content (*the fact that...*). By contrast, the acceptability of CP[*that*] complements (as in (1b)), which are prototypical vessels of propositions, exhibited gradience, ranging from fully acceptable (e.g., *annoyed*) to borderline unacceptable (e.g., *talk*). Since all predicates elicited high ratings in the PP condition, this gradience cannot be attributed to differences in *semantic* selectional restrictions. Moreover, as we used fully parallel materials across predicates, systematic processing differences are unlikely, and any such effects should have surfaced uniformly, including in the PP condition. The resulting *gradience* therefore points to gradient *syntactic* selectional restrictions: different predicates license CPs to different degrees. As syntactic selectional restrictions are part of the grammar, we conclude that *grammaticality – not just acceptability – is gradient*.

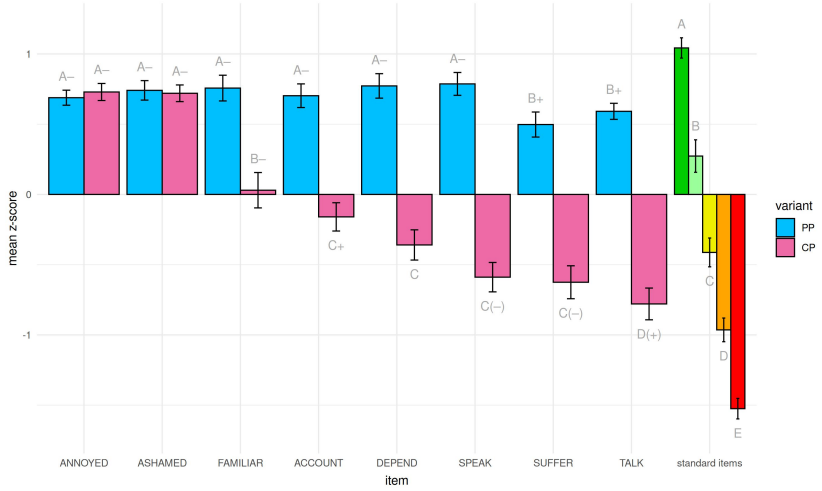
In Experiment 2, not reported here in detail due to space limitations, we replicated the mid-range CP acceptability for the six predicates exhibiting this pattern in Experiment 1 (i.e., without *annoyed* and *ashamed*), and we verified through modality tests that this is not a result of the existence of two groups of respondents: one accepting CP complements, and one rejecting them. (Some bimodality was observed only in the case of *speak*.) Also, in this and the following experiment, the sample was balanced for British and American English varieties, and the two groups responded similarly.

Finally, Experiment 3, conducted with 79 native speakers, tested whether the mid-range CP acceptability would extend to coordinate structures where the CP is a second conjunct, with the fully acceptable PP as the first (cf. Sag *et al.* 1985, as well as Bruening and Al Khalaf 2020, Patejuk and Przepiórkowski 2023, Bruening 2025, and Przepiórkowski and Patejuk 2025). This experiment used nine triples like (2a) (CP), (2b) (PP&CP), and (2c) (PP; note that in this case coordination is within the complement of the preposition, with the CP embedded in an NP) for selected four predicates under the same methodology; see Figure 1b for a summary of results.

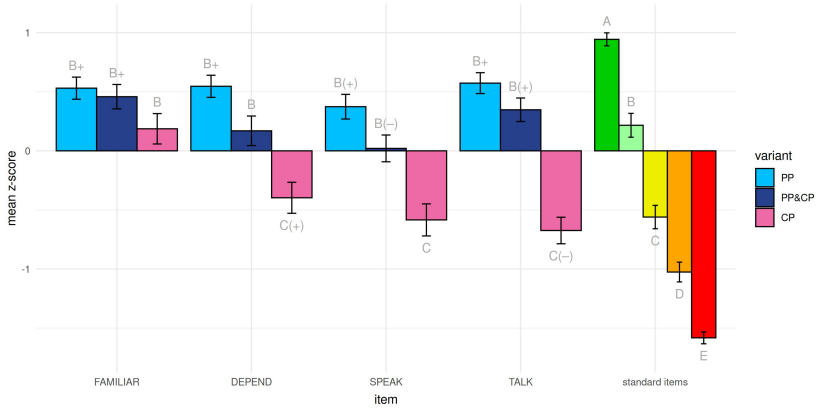
(2) a. Justin Bieber can **depend** [$_{\text{CP}}$ that [$_{\text{NP}}$ his fans]$_i$ still love his early songs].
b. Justin Bieber can **depend** [[$_{\text{PP}}$ on [$_{\text{NP}}$ his fans]$_i$] and [$_{\text{CP}}$ that they$_i$ still love his early songs]].
c. Justin Bieber can **depend** [$_{\text{PP}}$ on [[$_{\text{NP}}$ his fans]$_i$ and [$_{\text{NP}}$ the fact [$_{\text{CP}}$ that they$_i$ still love his early songs]]]].

Based on length and complexity alone, CP should be the most acceptable, followed by the PP&CP coordination, and finally PP. However, the observed differences are precisely the inverse, with PP complements the most acceptable and CP complements the least. Crucially, the acceptability of PP&CP was consistently (and in almost all cases significantly, $p < .05$) lower than that of PP, showing that syntactic selectional restrictions are also checked on second conjuncts, contrary to dominant theories of coordination (e.g., Munn 1993, Johannessen 1998, Zhang 2009, 2023).

In sum, we showed that syntactic selectional restrictions are gradient: predicates that fully license propositional PPs vary systematically in their willingness to take propositional CPs. Since syntactic selectional restrictions belong to the grammar, *grammaticality – not just acceptability – must be gradient*. An important corollary for research on human sentence processing is that not all gradience in acceptability judgements may be attributed to processing (or semantic) factors.



(a) Mean z-scores of stimuli and standard items in Experiment 1 (with 95% *CI*s)



(b) Mean z-scores of stimuli and standard items in Experiment 3 (with 95% *CI*s)

Figure 1: Results of Experiments 1 and 3 (about standard items occurring there, see Featherston 2008, 2020; in brief, A–B are grammatical, C is borderline, D–E are ungrammatical.)

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